Tutorial-3

A retail company wants to predict monthly sales revenue based on marketing and operational factors.

Input Features:

- Advertising spend (₹ in thousands)
- number of sales persons
- Store Football (per month)
- Discount rate (%)
- sales revenue (in thousand)

Target variables:

Ad spend	Salesperson	Football	Discount(%)	Sales revenue
50	5	1200	10	8.5
70	6	1500	12	10.2
40	4	1000	8	6.8
90	8	1900	15	14.0
60	6	1400	10	9.5
80	7	1700	14	12.5
55	5	1300	19	8.9
100	9	2100	18	16.0
65	6	1450	11	9.8
45	4	1100	8	7.2

1) Define the dependent variables

Dependent variable is monthly sales revenue because MSR depends on other factors like advertising spend, no. of Salesperson, store Football and discount.

2) write the multiple linear regression model

multiple linear regression model

$$y = B_0 + B_1 x_1 + B_2 x_2 + B_3 x_3 + B_4 x_4$$

where y = sales revenue

x_1 = Advertising spend

x_2 = no. of sales person

x_3 = ~~Foot~~ football store

x_4 = Discount rate

B_0 = intercept

B_1, B_2, B_3, B_4 = coefficients

3. Final model:

Sales = $\beta_0 + \beta_1$(Ad spend) + β_2(Sales person) + β_3(Football) + β_4(Discount)

substitute each rows

row1: $8.5 = \beta_0 + 50\beta_1 + 15\beta + 12000\beta_3 + 10\beta_4$

row2: $10.2 = \beta_0 + 70\beta_1 + 4\beta_2 + 15000\beta_3 + 12\beta_4$

row3: $6.8 = \beta_0 + 40\beta_1 + 4\beta_2 + 1000\beta_3 + 8\beta_4$

row4: $14 = \beta_0 + 90\beta_1 + 8\beta_2 + 1900\beta_3 + 15\beta_4$

row5: $9.5 = \beta_0 + 60\beta_1 + 6\beta_2 + 1400\beta_3 + 10\beta_4$

row6: $12.5 = \beta_0 + 80\beta_1 + 7\beta_2 + 1700\beta_3 + 24\beta_4$

row7: $8.9 = \beta_0 + 55\beta_1 + 5\beta_2 + 1300\beta_3 + 9\beta_4$

row8: $16.0 = \beta_0 + 100\beta_1 + 9\beta_2 + 2100\beta_3 + 18\beta_4$

row9: $9.8 = \beta_0 + 65\beta_1 + 6\beta_2 + 1400\beta_3 + 11\beta_4$

row10: $7.2 = \beta_0 + 45\beta_1 + 4\beta_2 + 1190\beta_3 + 8\beta_4$

Disscuss the effect of discount rate on sales growth revenue

Discount → 8, sales = 6.8; discount = 18, sales = 16, since 18 = 8 → 16 > 16.8

Sales increases with discount (positive effect)

Find MSE for 5 iterations:

$$\boxed{MSE = \frac{1}{N}\sum_{i=1}^{N}(y - \hat{y}_i)^2} \quad N = 10$$

$\hat{y}_1 = [8, 9, 7, 13, 10, 11, 9, 15, 10, 8]$

Square errors:

$(0.5)^2 + (1.2)^2 + (-0.2)^2 + 1^2 + (-0.5)^2 + (-0.1)^2 + 1^2 + (-0.2)^2 + (-0.8)^2$

$= 6.92 \Rightarrow MSE = \frac{6.92}{10} = 0.692$

Iteration: 2 $\hat{y}_2 = [8.3, 9.8, 6.9, 13.6, 9.7, 12, 8.8, 15.5, 9.9, 7.5]$

Sum of square error = 2.01

$MSE_2 = \frac{2.01}{10} = 0.201$

Iteration: 3 $\hat{y}_3 = [8.4, 10, 6, 13.8, 9.6, 12.3, 8.85, 15.3, 9.85, 7.35]$

$SSE = 0.62$

$MSE_3 = \frac{0.62}{10} = 0.062$

Iteration: 4 $\hat{y}_4 = [8.45, 10.1, 6.82, 13.9, 9.55, 12.4, 8.88, 15.9, 9.82, 7.28]$

$SSE = 0.18$

$MSE_4 = \frac{0.18}{10} = 0.018$

Iteration: 5 $\hat{y}_5 = [8.5, 10.21, 6.8, 14, 9.5, 12.5, 8.9, 16, 9.8, 7.2]$